Conserving mass (1)

When a chemical reaction occurs, the atoms in the reactants are rearranged to form the products. No new atoms are created, and none are destroyed. Mass is **conserved**. This means that the mass of the products at the end of the reaction is equal to the mass of the reactants at the beginning. We use **molecular drawings** to show this.

carbon C + oxygen gas O_2 ⟶ carbon dioxide CO_2



The number of atoms of each element at the start is the same as at the end, so we say that the equation is balanced. What if the equation is not balanced?

magnesium Mg + oxygen gas O_2 ⟶ magnesium oxide MgO



Something is wrong. There are more oxygen atoms at the start of the reaction than at the end, but we know that atoms cannot be lost during a chemical reaction. If there were two molecules of magnesium oxide at the end we would have the correct number of oxygen atoms, but one too many magnesium atoms. There must have been 2 magnesium atoms at the start of the reaction.



2Mg + O_2 **2**MgO

We say that this equation is balanced. We place a large number at the front to show how many molecules of each reactant and product there are. Placing a large number at the front of a formula multiplies the whole of the formula by that number. For example, **2MgO** means 2 atoms of magnesium **and** 2 atoms of oxygen.

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Conserving mass (2)

Read the Resource sheet 'Conserving mass (1)' then answer the following questions.

☞ **1** Below are two different chemical equations. Draw molecular diagrams to show how the atoms are rearranged and then decide if they are balanced or not. Remember to use different colours for different types of atoms.

a hydrogen carbonate ⟶ carbon dioxide + water

H_2CO_3 ⟶ CO_2 + H_2O

At the start there are	**At the end there are**
_____ hydrogen atoms	_____ hydrogen atoms
_____ carbon atoms	_____ carbon atoms
_____ oxygen atoms	_____ oxygen atoms
Total number of atoms =	Total number of atoms =

Is the equation balanced? If not, try increasing the number of different molecules. Remember to put a number at the front to show how many are needed.

b sodium + chlorine gas ⟶ sodium chloride

Na + Cl_2 ⟶ NaCl

At the start there are	**At the end there are**
_____ sodium atoms	_____ sodium atoms
_____ chlorine atoms	_____ chlorine atoms
Total number of atoms =	Total number of atoms =

Is the equation balanced? If not, try increasing the number of different molecules. Remember to put a number at the front to show how many are needed.

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